

MATH 225 – Applied Linear Algebra II

Course Notes

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Lecture 1: General Vector Spaces and Subspaces

Intuition: In previous courses, a "vector" was strictly an arrow in space (with magnitude and direction) or a column of numbers. In Applied Linear Algebra II, we generalize this concept entirely. A vector space is simply a mathematical sandbox. It is any collection of objects (whether they are matrices, polynomials, functions, or coordinates) that strictly obey a specific set of 10 rules. If they obey these rules, they are officially "vectors," and all the powerful theorems of linear algebra apply to them.

The first five axioms govern how addition works internally, and the next five govern how scaling interacts with the vectors.

Vector Addition Axioms

1. **Closure under Addition:** For all $u, v \in V$, the sum $u + v \in V$. (Adding two vectors cannot produce an object outside the space).
2. **Commutativity:** $u + v = v + u$ for all $u, v \in V$.
3. **Associativity:** $(u + v) + w = u + (v + w)$ for all $u, v, w \in V$.
4. **Additive Identity:** There exists a unique element $0 \in V$ such that $v + 0 = v$ for all $v \in V$.
5. **Additive Inverse:** For each $v \in V$, there exists a unique element $-v \in V$ such that $v + (-v) = 0$.

Scalar Multiplication Axioms

6. **Closure under Scalar Multiplication:** For all $c \in \mathbb{R}$ and $v \in V$, the product $cv \in V$.
7. **Vector Distributivity:** $c(u + v) = cu + cv$ for all $c \in \mathbb{R}$ and $u, v \in V$.
8. **Scalar Distributivity:** $(c + d)v = cv + dv$ for all $c, d \in \mathbb{R}$ and $v \in V$.
9. **Associativity:** $(cd)v = c(dv)$ for all $c, d \in \mathbb{R}$ and $v \in V$.
10. **Multiplicative Identity:** $1v = v$ for all $v \in V$.

The Elements of a Vector Space

The elements of a vector space V are called **vectors**. The elements of the field (which for this course is always $\mathbb{R}$) are called **scalars**.

Canonical Examples of Vector Spaces:

(1) **Euclidean Space $\mathbb{R}^n$:**

$$\mathbb{R}^n = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} : x_1, \dots, x_n \in \mathbb{R} \right\}$$

This is the standard space from linear algebra. Vector addition and scalar multiplication are performed component-wise. The zero vector is a column of n zeros.

(2) **Matrix Space $M_{m \times n}(\mathbb{R})$:** The set of all $m \times n$ matrices. Addition is the standard matrix addition (adding corresponding entries), and scalar multiplication scales every entry in the matrix. The additive identity here is the $m \times n$ zero matrix, consisting entirely of zeros.

(3) **Polynomial Space $P_n(\mathbb{R})$:** The space of all polynomials of degree $\leq n$. The elements look like $p(x) = a_0 + a_1x + \dots + a_nx^n$. It is defined with the operations:

$$(f + g)(x) = f(x) + g(x) \quad \text{and} \quad (cf)(x) = cf(x) \quad (c \in \mathbb{R}).$$

The zero vector in this space is the zero polynomial $z(x) = 0$ for all x . Under these operations, $P_n(\mathbb{R})$ is a valid vector space.

Fundamental Properties Derived from the Axioms

Intuition: Because we have redefined "vectors" to be anything from matrices to functions, we cannot blindly assume that basic arithmetic works the same way it does with regular numbers. We must formally prove that things like $0 \cdot \vec{x} = \vec{0}$ are guaranteed strictly by the 10 axioms, without relying on our intuition.

Theorem (Elementary Arithmetic Properties of Vector Spaces):

Let V be a vector space over $\mathbb{R}$. For all $\vec{x} \in V$ and all $t \in \mathbb{R}$:

1. $0\vec{x} = \vec{0}$
2. $t\vec{0} = \vec{0}$
3. $(-1)\vec{x} = -\vec{x}$

4. If $t\vec{x} = \vec{0}$, then either $t = 0$ or $\vec{x} = \vec{0}$.

Proof. Proof of (1): In the real numbers, $0 + 0 = 0$. By scalar distributivity (Axiom 8):

$$0\vec{x} = (0 + 0)\vec{x} = 0\vec{x} + 0\vec{x}.$$

Since $0\vec{x} \in V$ by closure, its additive inverse $-(0\vec{x})$ must exist in V (Axiom 5). Adding this inverse to both sides:

$$0\vec{x} + (-(0\vec{x})) = (0\vec{x} + 0\vec{x}) + (-(0\vec{x}))$$

$$\vec{0} = 0\vec{x} + (0\vec{x} + (-(0\vec{x}))) \quad (\text{Associativity})$$

$$\vec{0} = 0\vec{x} + \vec{0} \quad (\text{Additive Inverse})$$

$$\vec{0} = 0\vec{x} \quad (\text{Additive Identity}).$$

Therefore, $0\vec{x} = \vec{0}$.

Proof of (2): By the additive identity axiom, $\vec{0} + \vec{0} = \vec{0}$. Multiplying by scalar t and using vector distributivity (Axiom 7):

$$t\vec{0} = t(\vec{0} + \vec{0}) = t\vec{0} + t\vec{0}.$$

Adding $-(t\vec{0})$ to both sides and associating terms yields $t\vec{0} = \vec{0}$.

Proof of (3): We show that $(-1)\vec{x}$ behaves exactly as the unique additive inverse of $\vec{x}$. We add them together:

$$\vec{x} + (-1)\vec{x} = 1\vec{x} + (-1)\vec{x} = (1 + (-1))\vec{x} = 0\vec{x} = \vec{0}.$$

Since adding $(-1)\vec{x}$ to $\vec{x}$ results in the zero vector, it satisfies the definition of an inverse. By uniqueness of the additive inverse in V , $(-1)\vec{x} = -\vec{x}$.

Proof of (4): Suppose $t\vec{x} = \vec{0}$. If $t = 0$, the claim holds. If $t \neq 0$, the multiplicative inverse $t^{-1} \in \mathbb{R}$ exists. Multiplying both sides:

$$t^{-1}(t\vec{x}) = t^{-1}\vec{0} \implies (t^{-1}t)\vec{x} = \vec{0} \implies 1\vec{x} = \vec{0} \implies \vec{x} = \vec{0}.$$

□

Subspaces and the Subspace Test

Intuition: A subspace is simply a smaller, self-contained vector space living inside a larger one (for example, a 2D plane living inside a 3D space). Instead of verifying all 10 axioms to prove a subset is a vector space, we can take a shortcut. Since the subset inherits properties like commutativity and distributivity from the parent space, we only need to check if the subset is a "closed system." If it contains the origin, and doesn't spill out of its boundaries when we add or scale elements, it is a subspace.

Definition (Subspace).

Let V be a vector space over $\mathbb{R}$. A subset $U \subseteq V$ is called a **subspace** of V if U is itself a vector space over $\mathbb{R}$ under the addition and scalar multiplication operations inherited from V .

Theorem (Subspace Test).

Let V be a vector space over $\mathbb{R}$. A subset $U \subseteq V$ is a subspace of V if and only if:

1. $\vec{0} \in U$, where $\vec{0}$ is the zero vector of V (in particular, this proves $U \neq \emptyset$).
2. For all $\vec{x}, \vec{y} \in U$, $\vec{x} + \vec{y} \in U$ (**Closure under vector addition**).
3. For all $\alpha \in \mathbb{R}$ and $\vec{x} \in U$, $\alpha\vec{x} \in U$ (**Closure under scalar multiplication**).

Proof. ($\implies$) If U is a subspace, it satisfies all vector space axioms on its own. In particular, it must be closed under addition and scalar multiplication, and contain an additive identity $\vec{0}_U \in U$. Since $\vec{0}_U + \vec{0}_U = \vec{0}_U$, adding the inverse $-\vec{0}_U$ computed in V gives $\vec{0}_U = \vec{0}_V$.

($\impliedby$) Assume conditions (1), (2), and (3) hold. Closure under addition and scalar multiplication, as well as the existence of the additive identity, are directly satisfied by the assumptions. For any $\vec{x} \in U$, setting $\alpha = -1$ ensures that $-\vec{x} = (-1)\vec{x} \in U$. The remaining algebraic axioms (commutativity, associativity, and distributivity) hold for all elements in U automatically because they hold throughout the larger space V . Thus, U is a valid vector space. $\square$

Worked Examples: Subspace Analysis**Example 1 (Polynomial with Root Constraint).**

Let $V = P_2(\mathbb{R})$. Define the subset:

$$U = \{p(x) \in P_2(\mathbb{R}) \mid p(3) = 0\}.$$

Prove that U is a subspace of $P_2(\mathbb{R})$.

Proof. We test the three conditions of the Subspace Test systematically:

1. **Zero Vector:** The zero polynomial in $P_2(\mathbb{R})$ is $z(x) = 0 + 0x + 0x^2$. Evaluating at $x = 3$ gives $z(3) = 0$. Because it satisfies the condition $p(3) = 0$, we conclude $z \in U$.
2. **Closure under Addition:** Let $p(x), q(x) \in U$. By definition, this means $p(3) = 0$ and $q(3) = 0$. Consider their sum evaluated at $x = 3$:

$$(p + q)(3) = p(3) + q(3) = 0 + 0 = 0.$$

Since the sum evaluated at 3 is zero, $p + q \in U$.

3. **Closure under Scalar Multiplication:** Let $p(x) \in U$ and $\alpha \in \mathbb{R}$. Then:

$$(\alpha p)(3) = \alpha \cdot p(3) = \alpha(0) = 0.$$

Thus, $\alpha p \in U$.

By the Subspace Test, all conditions hold, so U is a subspace of $P_2(\mathbb{R})$. $\square$

Example 2 (Traceless Matrices).

Let $V = M_{2 \times 2}(\mathbb{R})$. Define:

$$W = \{A \in M_{2 \times 2}(\mathbb{R}) \mid \text{tr}(A) = 0\}.$$

Recall that for $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the trace is the sum of the main diagonal $\text{tr}(A) = a + d$. Show that W is a subspace of $M_{2 \times 2}(\mathbb{R})$.

Proof. 1. **Zero Vector:** The zero matrix $O_{2 \times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ satisfies $\text{tr}(O_{2 \times 2}) = 0 + 0 = 0$. Hence,
 $O_{2 \times 2} \in W$.

2. **Closure under Addition:** Let $A, B \in W$, meaning $\text{tr}(A) = 0$ and $\text{tr}(B) = 0$. Because the trace operator is linear over matrix addition:

$$\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B) = 0 + 0 = 0.$$

Hence, $A + B \in W$.

3. **Closure under Scalar Multiplication:** Let $A \in W$ and $\alpha \in \mathbb{R}$. Again, trace is linear over scalar multiplication:

$$\text{tr}(\alpha A) = \alpha \text{tr}(A) = \alpha(0) = 0.$$

Hence, $\alpha A \in W$.

Thus, W is a subspace of $M_{2 \times 2}(\mathbb{R})$. $\square$

Example 3 (Affine Coefficient Failure).

Let $V = P_2(\mathbb{R})$ and consider the subset:

$$S = \{p(x) = ax^2 + bx + c \in P_2(\mathbb{R}) \mid a - 2b + c = 3\}.$$

Determine whether S is a subspace of V .

Solution:

We always check the zero vector first. The zero polynomial $z(x) = 0x^2 + 0x + 0$ has coefficients $a = 0, b = 0, c = 0$. Evaluating the subset's defining equation with these coefficients:

$$a - 2b + c = 0 - 2(0) + 0 = 0 \neq 3.$$

Because the zero polynomial does not satisfy the condition, it is not in S . Condition (1) of the Subspace Test fails immediately. A subspace *must* always contain the zero vector.

Geometrically, in the "space" of coefficients, this represents a plane that has been shifted away from the origin by 3 units. Thus, S is **not** a subspace of $P_2(\mathbb{R})$.

Example 4 (A Plane Through the Origin in $\mathbb{R}^3$).

Let $V = \mathbb{R}^3$. Consider the subset U representing a plane passing through the origin:

$$U = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \mid x - 2y + 3z = 0 \right\}$$

Prove that U is a subspace of $\mathbb{R}^3$.

Proof. 1. **Zero Vector:** The zero vector $\vec{0} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$. Evaluating the condition: $0 - 2(0) + 3(0) = 0$.

Thus, $\vec{0} \in U$.

2. **Closure under Addition:** Let $\vec{u} = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$ be in U . This implies $x_1 - 2y_1 + 3z_1 = 0$ and $x_2 - 2y_2 + 3z_2 = 0$.

Consider their sum $\vec{u} + \vec{v} = \begin{bmatrix} x_1 + x_2 \\ y_1 + y_2 \\ z_1 + z_2 \end{bmatrix}$. We test the condition:

$$(x_1 + x_2) - 2(y_1 + y_2) + 3(z_1 + z_2) = (x_1 - 2y_1 + 3z_1) + (x_2 - 2y_2 + 3z_2) = 0 + 0 = 0.$$

Thus, $\vec{u} + \vec{v} \in U$.

3. **Closure under Scalar Multiplication:** Let $\vec{u} \in U$ and $\alpha \in \mathbb{R}$. The scaled vector is $\alpha\vec{u} = \begin{bmatrix} \alpha x_1 \\ \alpha y_1 \\ \alpha z_1 \end{bmatrix}$. Testing the condition:

$$(\alpha x_1) - 2(\alpha y_1) + 3(\alpha z_1) = \alpha(x_1 - 2y_1 + 3z_1) = \alpha(0) = 0.$$

Thus, $\alpha\vec{u} \in U$.

Since it passes the Subspace Test, the plane U is a subspace of $\mathbb{R}^3$. □

Lecture 2: Span and Linear Independence

Intuition: If we treat vectors like directional building blocks, the "Span" is the set of absolutely everywhere you can possibly reach by combining them. "Linear Independence" means there is no redundancy in your building blocks; no vector in the set can be created by mixing the others. If a set spans a space and is linearly independent, it forms a perfect, highly efficient coordinate system for that space.

Linear Span and Minimal Subspaces

Definition (Linear Combination and Span).

Let V be a vector space over $\mathbb{R}$ and let $S = \{\vec{v}_1, \dots, \vec{v}_k\} \subseteq V$.

1. An expression of the form $c_1\vec{v}_1 + \dots + c_k\vec{v}_k$ with $c_i \in \mathbb{R}$ is called a **linear combination** of the vectors in S .
2. The **span** of S , denoted $\text{Span}(S)$, is the set of all possible linear combinations:

$$\text{Span}(S) = \{c_1\vec{v}_1 + \dots + c_k\vec{v}_k \mid c_1, \dots, c_k \in \mathbb{R}\}.$$

If $S = \emptyset$, we define $\text{Span}(\emptyset) = \{\vec{0}\}$.

Theorem: (Span is a Subspace).

Let V be a vector space over $\mathbb{R}$ and let $S = \{\vec{v}_1, \dots, \vec{v}_k\} \subseteq V$. Then $\text{Span}(S)$ is a subspace of V .

Proof. We verify the three conditions:

1. **Zero Vector:** Choosing $c_1 = \dots = c_k = 0$, we have $\vec{0} = 0\vec{v}_1 + \dots + 0\vec{v}_k \in \text{Span}(S)$.
2. **Closure under Addition:** Let $\vec{x}, \vec{y} \in \text{Span}(S)$. There exist scalars $c_i, d_i \in \mathbb{R}$ such that:

$$\vec{x} = \sum_{i=1}^k c_i \vec{v}_i \quad \text{and} \quad \vec{y} = \sum_{i=1}^k d_i \vec{v}_i.$$

Adding these vectors and applying distributivity:

$$\vec{x} + \vec{y} = \sum_{i=1}^k c_i \vec{v}_i + \sum_{i=1}^k d_i \vec{v}_i = \sum_{i=1}^k (c_i + d_i) \vec{v}_i.$$

Since each $c_i + d_i \in \mathbb{R}$, $\vec{x} + \vec{y} \in \text{Span}(S)$.

3. **Closure under Scalar Multiplication:** For any scalar $\alpha \in \mathbb{R}$:

$$\alpha \vec{x} = \alpha \left(\sum_{i=1}^k c_i \vec{v}_i \right) = \sum_{i=1}^k (\alpha c_i) \vec{v}_i.$$

Since each $\alpha c_i \in \mathbb{R}$, $\alpha \vec{x} \in \text{Span}(S)$.

□

Worked Examples: Proving a Span

Intuition for proving a span: To prove that a set of vectors S spans an entire vector space V , you must show that an *arbitrary* (generic) vector in V can always be built using a linear combination of the vectors in S . This always reduces to proving that a system of linear equations is consistent (has a solution) for any possible inputs.

Example 1 (Spanning $\mathbb{R}^2$).

Determine whether the set $S = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$ spans $\mathbb{R}^2$.

Solution. Let $\vec{w} = \begin{bmatrix} x \\ y \end{bmatrix}$ be an arbitrary vector in $\mathbb{R}^2$. We must determine if there always exist scalars $c_1, c_2 \in \mathbb{R}$ such that:

$$c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

This vector equation produces a system of two linear equations:

$$c_1 + c_2 = x \quad (\text{Equation 1})$$

$$c_1 - c_2 = y \quad (\text{Equation 2})$$

We solve for c_1 and c_2 in terms of x and y . Adding the two equations directly eliminates c_2 :

$$(c_1 + c_2) + (c_1 - c_2) = x + y \implies 2c_1 = x + y \implies c_1 = \frac{x + y}{2}$$

Subtracting Equation 2 from Equation 1 eliminates c_1 :

$$(c_1 + c_2) - (c_1 - c_2) = x - y \implies 2c_2 = x - y \implies c_2 = \frac{x - y}{2}$$

Because we can find explicit, real values for c_1 and c_2 regardless of what x and y are, every vector in $\mathbb{R}^2$ can be constructed. Therefore, S spans $\mathbb{R}^2$. □

Linear Independence and Unique Representation

Definition (Linear Dependence and Independence).

Let V be a vector space over $\mathbb{R}$ and let $S = \{\vec{v}_1, \dots, \vec{v}_k\} \subseteq V$.

- The set S is **linearly dependent** if there exist scalars $c_1, \dots, c_k \in \mathbb{R}$, **not all zero**, such that:

$$c_1\vec{v}_1 + c_2\vec{v}_2 + \dots + c_k\vec{v}_k = \vec{0}.$$

- The set S is **linearly independent** if the only solution to

$$c_1\vec{v}_1 + c_2\vec{v}_2 + \dots + c_k\vec{v}_k = \vec{0}$$

is the trivial solution: $c_1 = c_2 = \dots = c_k = 0$.

Worked Examples: Checking Linear Independence

Intuition for testing independence: To test for linear independence, we always set a generic linear combination of the set equal to the zero vector. We then solve for the coefficients c_i .

- If we get a unique solution ($c_1 = c_2 = \dots = 0$), the set is **independent**.
- If we get free variables (meaning infinitely many non-zero solutions exist), the set is **dependent**, and one vector is just a combination of the others.

Example 2 (Linear Dependence in $\mathbb{R}^3$).

Determine whether the set $S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 7 \\ 1 \end{bmatrix} \right\}$ is linearly independent.

Solution. We set up the dependence equation:

$$c_1 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 3 \\ 7 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

We translate this vector equation into an augmented matrix and row reduce:

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 2 & 5 & 7 & 0 \\ 1 & 0 & 1 & 0 \end{array} \right) \xrightarrow{\substack{R_2 \leftarrow R_2 - 2R_1 \\ R_3 \leftarrow R_3 - R_1}} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & -2 & -2 & 0 \end{array} \right) \xrightarrow{R_3 \leftarrow R_3 + 2R_2} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Because the third column lacks a leading 1 (a pivot), c_3 is a free variable. This means there are infinitely many non-zero solutions. For instance, if we let the free variable $c_3 = -1$, then from the second row ($c_2 + c_3 = 0$) we get $c_2 = 1$. From the first row ($c_1 + 2c_2 + 3c_3 = 0$), we get $c_1 + 2(1) + 3(-1) = 0 \implies c_1 = 1$. Since $1\vec{v}_1 + 1\vec{v}_2 - 1\vec{v}_3 = \vec{0}$ (a non-trivial combination equals zero), the set is **linearly dependent**. $\square$

Example 3 (Linear Independence in Polynomial Space).

Determine whether the set of polynomials $S = \{1 + x, x + x^2, 1 + x^2\}$ is linearly independent in $P_2(\mathbb{R})$.

Solution. Set a linear combination of the polynomials equal to the zero polynomial:

$$c_1(1 + x) + c_2(x + x^2) + c_3(1 + x^2) = 0 + 0x + 0x^2$$

Distribute the scalars and group the terms by powers of x :

$$(c_1 + c_3) + (c_1 + c_2)x + (c_2 + c_3)x^2 = 0 + 0x + 0x^2$$

For polynomials to be identically equal, their corresponding coefficients must match. This gives a system of three equations:

$$c_1 + c_3 = 0 \quad (\text{Constants})$$

$$c_1 + c_2 = 0 \quad (x \text{ coefficients})$$

$$c_2 + c_3 = 0 \quad (x^2 \text{ coefficients})$$

From the first equation, $c_3 = -c_1$. From the second equation, $c_2 = -c_1$. Substitute these into the third equation:

$$(-c_1) + (-c_1) = 0 \implies -2c_1 = 0 \implies c_1 = 0.$$

If $c_1 = 0$, then $c_2 = 0$ and $c_3 = 0$. Because the trivial solution is the unique solution to the system, the set of polynomials is **linearly independent**. $\square$

The Importance of Independence

Theorem (Unique Representation Theorem).

Let V be a vector space over $\mathbb{R}$ and let $S = \{\vec{v}_1, \dots, \vec{v}_k\} \subseteq V$ be a spanning set for V . Then every vector in V can be uniquely expressed as a linear combination of the vectors in S if and only if S is linearly independent.

Proof. ($\implies$) Suppose every vector has a unique representation. Since $\vec{0} \in V$, it can be written as:

$$0\vec{v}_1 + \dots + 0\vec{v}_k = \vec{0}.$$

Suppose there exist scalars $c_1, \dots, c_k$ such that $\sum_{i=1}^k c_i \vec{v}_i = \vec{0}$. By uniqueness of representation for $\vec{0}$, we must have $c_1 = \dots = c_k = 0$. Hence, S is linearly independent.

($\Leftarrow$) Suppose S is linearly independent. Let $\vec{v} \in V$. Because S spans V , there exists at least one representation $\vec{v} = \sum_{i=1}^k c_i \vec{v}_i$. Suppose there exists another representation $\vec{v} = \sum_{i=1}^k d_i \vec{v}_i$. Subtracting the two gives:

$$\sum_{i=1}^k c_i \vec{v}_i - \sum_{i=1}^k d_i \vec{v}_i = \vec{0} \implies \sum_{i=1}^k (c_i - d_i) \vec{v}_i = \vec{0}.$$

Because S is linearly independent, every scalar coefficient attached to the vectors must be exactly zero:

$$c_i - d_i = 0 \implies c_i = d_i \quad \text{for all } i = 1, \dots, k.$$

Because the coefficients must be identical, the linear combination is unique. □

Lecture 3: Bases, Dimension, and Expansion

Intuition: A basis is the perfect skeleton of a vector space. It is a set of vectors that spans the entire space (so you can reach absolutely everywhere) but is also linearly independent (so there is absolutely no redundancy in your coordinates). Because of these two properties, a basis provides a perfect, unique coordinate system for any space, ensuring every single vector has exactly one "address."

Bases and Standard Bases

Definition (Basis).

Let V be a vector space. A subset $\mathcal{B} = \{\vec{v}_1, \dots, \vec{v}_k\} \subseteq V$ is called a **basis** for V if it satisfies two conditions:

1. $\mathcal{B}$ is linearly independent (no redundant vectors).
2. $\text{Span}(\mathcal{B}) = V$ (enough vectors to build the whole space).

Convention: The basis for the trivial subspace $\{\vec{0}\}$ is the empty set $\emptyset$.

Standard Bases for Canonical Spaces

Just like the x, y, z axes in geometry, our abstract vector spaces have "standard" bases that are the simplest to work with:

- **Euclidean Space $\mathbb{R}^3$:** The standard basis consists of the unit coordinate vectors:

$$\mathcal{B} = \{\vec{e}_1, \vec{e}_2, \vec{e}_3\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

- **Matrix Space $M_{2 \times 2}(\mathbb{R})$:** The standard basis consists of single-entry unit matrices, one for each position:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} = \{E_{11}, E_{12}, E_{21}, E_{22}\}.$$

- **Polynomial Space $P_2(\mathbb{R})$:** The standard basis is the monomial basis, separating the constant, linear, and quadratic terms: $\mathcal{B} = \{1, x, x^2\}$.

Verification that $\{1, x, x^2\}$ is a Basis for $P_2(\mathbb{R})$. To rigorously prove this is a basis, we must check both criteria in the definition.

1. **Spanning:** Let $p(x) \in P_2(\mathbb{R})$ be an arbitrary polynomial. By definition, it has the form $p(x) = a + bx + cx^2$ where $a, b, c \in \mathbb{R}$. We can write this exactly as a linear combination of our basis elements:

$$p(x) = a(1) + b(x) + c(x^2).$$

Since any $p(x)$ can be constructed this way, the set spans the space: $P_2(\mathbb{R}) = \text{Span}\{1, x, x^2\}$.

2. **Linear Independence:** Suppose a linear combination equals the zero polynomial, $0(x)$, which evaluates to 0 for all $x \in \mathbb{R}$:

$$c_1(1) + c_2(x) + c_3(x^2) = 0(x)$$

Because this equation is an identity (it must hold true for *any* real number x), we can strategically plug in values for x to solve for the coefficients.

Evaluating at $x = 0$ kills the x and x^2 terms, giving $c_1 = 0$.

Evaluating at $x = 1$ and $x = -1$ yields a small system:

$$c_2 + c_3 = 0 \quad \text{and} \quad -c_2 + c_3 = 0 \implies 2c_3 = 0 \implies c_3 = 0.$$

Substituting $c_3 = 0$ back in gives $c_2 = 0$.

Thus, $c_1 = c_2 = c_3 = 0$. Because the trivial solution is the only solution, $\{1, x, x^2\}$ is linearly independent.

Because $\{1, x, x^2\}$ spans $P_2(\mathbb{R})$ and is linearly independent, it is a basis for $P_2(\mathbb{R})$. □

Theorem (Removing Redundant Vectors):

Let V be a non-empty vector space and suppose $\vec{v}_1, \dots, \vec{v}_k \in V$. If a vector $\vec{v}_i$ is a linear combination of the others, it is essentially "dead weight." Removing it does not alter the span of the set:

$$\text{Span}\{\vec{v}_1, \dots, \vec{v}_k\} = \text{Span}\{\vec{v}_1, \dots, \vec{v}_{i-1}, \vec{v}_{i+1}, \dots, \vec{v}_k\}.$$

Worked Examples: Finding a Basis

Example 1:

$$\text{Let } U = \text{Span} \left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} -2 \\ -2 \\ 2 \\ -6 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 0 \\ 5 \end{bmatrix} \right\}. \text{ Find a basis for } U.$$

Solution. To find a basis, we must filter out the redundant vectors. We first form a matrix whose columns are the given vectors. By row reducing this matrix, we track the dependencies between

the columns.

$$\begin{bmatrix} 1 & -2 & 2 & 3 \\ 1 & -2 & 1 & 2 \\ -1 & 2 & 1 & 0 \\ 3 & -6 & 2 & 5 \end{bmatrix}$$

We perform row reduction to find an REF (and its reduced form, RREF). First, we use the leading 1 in the first column to eliminate all entries below it:

$$\xrightarrow{\begin{matrix} R_2 \leftarrow R_2 - R_1 \\ R_3 \leftarrow R_3 + R_1 \\ R_4 \leftarrow R_4 - 3R_1 \end{matrix}} \begin{bmatrix} 1 & -2 & 2 & 3 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & -4 & -4 \end{bmatrix}$$

Notice that the second column does not have a pivot (a leading non-zero entry). We move to the third column to establish our next pivot by multiplying row 2 by -1 :

$$\xrightarrow{R_2 \leftarrow -R_2} \begin{bmatrix} 1 & -2 & 2 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & -4 & -4 \end{bmatrix}$$

Now, we use this new pivot in the third column to eliminate the entries below it:

$$\xrightarrow{\begin{matrix} R_3 \leftarrow R_3 - 3R_2 \\ R_4 \leftarrow R_4 + 4R_2 \end{matrix}} \begin{bmatrix} 1 & -2 & 2 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Finally, we place the matrix into full Reduced Row Echelon Form (RREF) by clearing the entry above the pivot in column 3:

$$\xrightarrow{R_1 \leftarrow R_1 - 2R_2} \begin{bmatrix} 1 & -2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The leading entries (the pivots) are located in **column 1** and **column 3**. This tells us that columns 2 and 4 are free columns, meaning they are linear combinations of columns 1 and 3 (they are redundant). To form our basis, we discard the redundant columns and keep the **original vectors** corresponding to the pivot columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix} \right\}$$

□

Example 2:

Let $U = \text{Span}\{2 - x + x^2, 1 + x - 2x^2, 3 - x^2\}$. Find a basis for U .

Solution. We must check these polynomials for linear independence. Suppose $c_1, c_2, c_3 \in \mathbb{R}$ such that their linear combination equals the zero polynomial:

$$c_1(2 - x + x^2) + c_2(1 + x - 2x^2) + c_3(3 - x^2) = 0.$$

If we expand and group the terms by their powers of x (constants together, x terms together, x^2 terms together), we force their coefficients to equal 0. This gives us the following **Linear System**:

$$2c_1 + c_2 + 3c_3 = 0 \quad (\text{constants})$$

$$-c_1 + c_2 = 0 \quad (x \text{ terms})$$

$$c_1 - 2c_2 - c_3 = 0 \quad (x^2 \text{ terms})$$

We extract the coefficients into a matrix and perform row reduction to find the pivots:

$$\xrightarrow{\text{Coefficient Matrix}} \begin{bmatrix} 2 & 1 & 3 \\ -1 & 1 & 0 \\ 1 & -2 & -1 \end{bmatrix} \xrightarrow{\text{REF}} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

The leading entries (pivots) appear in the first and second columns. This means the third column represents a redundant polynomial. Thus, we extract the first two original polynomials to form our basis:

$$\mathcal{B} = \{2 - x + x^2, 1 + x - 2x^2\}$$

□

Dimension of a Vector Space

If $B = \{\vec{v}_1, \dots, \vec{v}_k\}$ and $C = \{\vec{w}_1, \dots, \vec{w}_l\}$ are both bases for a vector space V , then it is a mathematical guarantee that $k = l$. A vector space always requires the exact same number of basis vectors, regardless of which specific vectors you choose.

Definition (Dimension).

If $B = \{\vec{v}_1, \dots, \vec{v}_k\}$ is a basis for a vector space V , then the **dimension** of V , denoted by $\dim(V)$, is the cardinality (size) of B . Written as:

$$\dim(V) = k \quad \text{In particular, if } V = \{\vec{0}\}, \text{ then } \dim(V) = 0.$$

Examples of Dimension:

1. $\dim(\mathbb{R}^n) = n$ (requires exactly n coordinate vectors).
2. $\dim(M_{m \times n}(\mathbb{R})) = m \times n$ (requires one vector for every matrix entry).
3. $\dim(P_n(\mathbb{R})) = n + 1$ (the $+1$ is for the constant term).

Theorem:

Let V be a finite dimensional space and suppose that $\dim(V) = n$. Then:

1. A set of **more than** n vectors must be linearly dependent. (You have too many vectors, creating guaranteed redundancy).
2. A set of **fewer than** n vectors cannot span V . (You don't have enough directions to reach every corner of the space).
3. A set of **exactly** n vectors in V spans V if and only if it is linearly independent. (If the count is perfectly n , passing one test automatically guarantees passing the other).

Examples:

- Let $B = \{1, x, x^2, 2x^2, -3x + 1\} \subseteq P_2(\mathbb{R})$. The set B is linearly dependent because it contains 5 elements, which is more than $\dim(P_2(\mathbb{R})) = 3$. Because $5 > 3$, there is unavoidable redundancy. Therefore, B cannot be a basis.

- Let $B = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\} \subseteq \mathbb{R}^3$. Because there are only 2 vectors and $\dim(\mathbb{R}^3) = 3$, this set physically cannot span $\mathbb{R}^3$. It only describes a 2D plane.

Basis Expansion Theorem

Theorem (Basis Expansion):

Let V be an n -dimensional vector space. If $C = \{\vec{v}_1, \dots, \vec{v}_k\} \subseteq V$ is a linearly independent set with $k < n$, then we can find additional vectors $\vec{v}_{k+1}, \dots, \vec{v}_n \in V$ such that:

$$\{\vec{v}_1, \dots, \vec{v}_k, \vec{v}_{k+1}, \dots, \vec{v}_n\}$$

is a full basis for V . (In other words, you can always "pad" an independent set until it becomes a basis).

Steps to expand a basis:

1. Consider the independent set.
2. Form a matrix with a generic vector and find an REF to determine the dependency conditions.
3. Choose a new vector that intentionally violates those conditions.

Example 1 (Expanding in $\mathbb{R}^3$):

Extend the set $C = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$ to a basis for $\mathbb{R}^3$.

Solution. We note that C is linearly independent (they are not scalar multiples). Since $\dim(\mathbb{R}^3) = 3$, we need to add exactly one more vector to C , ensuring it is NOT a linear combination of the vectors currently in C .

Let $\begin{bmatrix} a \\ b \\ c \end{bmatrix} \in \mathbb{R}^3$ be a generic vector. We check the condition for this vector to fall inside the span

by checking if there exist $c_1, c_2 \in \mathbb{R}$ such that:

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Setting up the augmented matrix of this system and computing the REF, we get:

$$\left(\begin{array}{cc|c} 1 & 0 & a \\ 0 & 1 & b - 2a \\ 0 & 0 & c - 3a \end{array} \right)$$

Looking at the bottom row, this system is inconsistent (meaning the vector is outside the span) when $c - 3a \neq 0 \implies c \neq 3a$. Choosing any numbers for a, b, c that deliberately break this rule will give us a valid extension vector.

Consider $a = 2, b = 0, c = 1$. Because $1 \neq 3(2)$, the rule is broken, so the vector $\begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$ is

completely independent from the original two.

So, expanding the set gives:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

which is a full basis for $\mathbb{R}^3$. □

Example 2 (Expanding in Polynomial Space):

Extend the set $C = \{x, 1 + x\}$ to a basis for $P_3(\mathbb{R})$.

Solution. The space $P_3(\mathbb{R})$ is 4-dimensional, so we need two more polynomials that are not in the span of C . Notice that C contains only degree 1 polynomials. Any combination of C will only yield degree 1 or constant terms; it physically cannot create any polynomials of degree 2 or higher.

Because of this, we can safely and simply add the standard higher-degree polynomials x^2 and x^3 . Since they cannot be built from C , they maintain linear independence:

$$\mathcal{B} = \{x, 1 + x, x^2, x^3\}$$

□